



# Tim van Beeck

Ph.D. Student · Numerical mathematics

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## Scientific biography

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I am a Ph.D. student in Applied Mathematics at the University of Göttingen, supervised by Christoph Lehrenfeld. My research focuses on the numerical analysis of partial differential equations. I am currently working on the discretization of time-harmonic wave propagation problems arising in nematoacoustics and helioseismology.

## Education

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| since 04/2024     | <b>Ph.D. in Mathematics</b> , University of Göttingen.<br><i>Supervisor:</i> Christoph Lehrenfeld.                                                                                                                           |
| 12/2021 - 12/2023 | <b>M.Sc. in Mathematics</b> , University of Göttingen, Grade: 1.0 (with distinction).<br><i>Thesis:</i> On stable discontinuous Galerkin discretizations for Galbrun's equation.<br><i>Supervisor:</i> Christoph Lehrenfeld. |
| 10/2017 - 12/2021 | <b>B.Sc. in Mathematics</b> , University of Göttingen, Grade: 2.3.<br><i>Thesis:</i> On a Discontinuous Galerkin discretization for a degenerate diffusion equation.<br><i>Supervisor:</i> Christoph Lehrenfeld.             |
| 2009 - 2017       | <b>Secondary school</b> (Abitur), Gymnasium Corvinianum Northeim.                                                                                                                                                            |

## Experience

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### Positions:

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| since 01/2024     | <b>Research assistant</b> , University of Göttingen, Germany. |
| 11/2021 - 12/2023 | <b>Student assistant</b> , University of Göttingen, Germany.  |

### Teaching:

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|-------------------|--------------------------------------------------------------------------------------------------|
| 05/2026 - now     | Mathematically oriented programming (University of Göttingen, Lecture and tutorial class, 4SWS). |
| 10/2025 - 03/2026 | Mathematically oriented programming (University of Göttingen, tutorial class, 2SWS).             |
| 10/2023 - 03/2024 | Partial differential equations (University of Göttingen, tutorial class, 2SWS).                  |

## Publications

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### In preparation:

- [1] J. Preuss, P. Stocker, **T. van Beeck**, *Learned infinite elements for the vector-valued Galbrun's equation.*
- [2] **T. van Beeck**, *A  $C^0$ -hybrid interior penalty method for the nematic Helmholtz–Korteweg equation.*

### Preprints:

- [1] T. Brüers, C. Lehrenfeld, **T. van Beeck**, M. Wardetzky, *High-order surface flow discretizations via discrete Helmholtz-Hodge decompositions*. arXiv:2603.27714
- [2] P. L. Lederer, C. Lehrenfeld, C. Merdon, **T. van Beeck**, *Pressure-robustness for the axisymmetric Stokes problem by velocity reconstruction*. arXiv:2603.01030.
- [3] P. E. Farrell, **T. van Beeck**, U. Zerbinati, *Analysis and numerical analysis of the Helmholtz–Korteweg equation*. arXiv:2503.10771.
- [4] **T. van Beeck**, U. Zerbinati, *An adaptive mesh refinement strategy to ensure quasi-optimality of finite element methods for self-adjoint Helmholtz problems*. arXiv:2403.06266.

#### Journal articles:

- [1] C. Lehrenfeld, **T. van Beeck**, I. Voulis, *Analysis of divergence-preserving unfitted finite element methods for the mixed Poisson problem*. Math. Comp., Vol. 94 (2025), 1667-1699, DOI: 10.1090/mcom/4027.
- [2] M. Halla, C. Lehrenfeld, **T. van Beeck**, *Hybrid discontinuous Galerkin discretizations for the damped time-harmonic Galbrun’s equation*. ESAIM: M2AN, Vol. 60 (2026), 1135 - 1175, DOI: 10.1051/m2an/2026027.
- [3] E. Burman, J. Preuss, **T. van Beeck**, *Variational data assimilation for the wave equation in heterogeneous media*. Accepted in Commun. Appl. Math. Comput., arXiv:2509.13108.

## Scientific exchange

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#### Research visits:

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|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 03/2026           | University of Oxford, United Kingdom, working with U. Zerbinati & P. E. Farrell.                                                                            |
| 10/2025           | University of Oxford, United Kingdom, working with U. Zerbinati & P. E. Farrell.                                                                            |
| 12/2024           | Pavia, Italy, working with U. Zerbinati.                                                                                                                    |
| 01/2024 - 03/2024 | UCL London, United Kingdom, working with E. Burman & J. Preuss<br><i>Topic: Variational data assimilation for the wave equation in heterogeneous media.</i> |

#### Conference talks:

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|---------|--------------------------------------------------------------------------------------------------------------------------|
| 09/2025 | ENUMATH 2025, Heidelberg, Germany, <i>Divergence-preserving unfitted FEM for the mixed Poisson problem</i> .             |
| 05/2025 | EFEF 2025, Trieste, Italy, <i>Pressure-robustness in an axisymmetric setting</i> .                                       |
| 02/2025 | Wave Phenomena 2025, Karlsruhe, Germany, <i>Learned infinite elements for the vector-valued Galbrun’s equation</i> .     |
| 11/2024 | GAMM Numerical Analysis, Augsburg, Germany, <i>Hybrid discontinuous Galerkin discretizations of Galbrun’s equation</i> . |
| 09/2024 | FEM Symposium 2024, Chemnitz, Germany, <i>Divergence-preserving unfitted FEM for the mixed Poisson problem</i> .         |
| 07/2024 | WCCM 2024, Vancouver, Canada, <i>Divergence-preserving unfitted FEM for the mixed Poisson problem</i> .                  |
| 06/2024 | WAVES 2024, Berlin, Germany, <i>Stable hybrid discontinuous Galerkin discretization for Galbrun’s equation</i> .         |
| 06/2024 | EFEF 2024, London, United Kingdom, <i>Ensuring quasi-optimality for the Helmholtz problem</i> .                          |
| 05/2023 | EFEF 2023, Enschede, Netherlands, <i>Unfitted Mixed Finite Element Methods</i> .                                         |
| 06/2022 | 41st NoKo, Hannover, Germany, <i>Unfitted Mixed Finite Element Methods</i> .                                             |

#### Community engagement:

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| 05/2025 - now | GAMM Junior Mentoring Programm (w. Benedikt Gräßle) |
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## Administrative work

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|-------------------|------------------------------------------------------------------------------------------------------------|
| 04/2024 - 03/2027 | Managing board of the institute for numerical and applied mathematics (representing scientific employees). |
| 04/2025 - 03/2026 | Deputy PhD students representative (Faculty of Computer Science & Mathematics).                            |